

Vansha Hansda

UNDERGRAD PHYSICS STUDENT

22b1803@iitb.ac.in | <https://github.com/vansha1>

TECHNICAL SKILLS

- **Programming Languages** : HTML, CSS, JavaScript, Python, Java, C++, C, Latex
- **Libraries** : pandas, NumPy, matplotlib, SciPy, rasterio, scikit-learn, seaborn, Numba
- **Software** : Canva, Figma, AutoDesk Fusion 360, Arduino IDE, MATLAB, DS9, Aperture Photometry Tool

PROJECTS

Crop Yield Estimation using LANDSAT Data

Aug 2025 - Nov 2025

Course Project | Digital Image Processing for Remotely Sensed Data | [presentation](#)

- Built a **Python pipeline** to convert raw **multi-band data from LANDSAT** into **structured datasets**
- Implemented **data preprocessing** and **feature engineering** for vegetation index computation
- Applied algorithmic techniques such as **PCA** and **clustering** to identify patterns in seasonal crop behaviour
- Developed a **predictive classification workflow** to infer crop performance from early-season satellite signals

Personal Website

May 2025 - July 2025

Self Project | **HTML, CSS** | [website link](#)

- Designed and coded a **portfolio website from scratch** to showcase creative work, writing and visuals
- Implemented **responsive layout** and **smooth scrolling**, with images, text overlays, and custom typography
- **Deployed** a **live website** on **GitHub Pages** as digital scrapbook to present professional and personal projects

Coffee Shop Game

May 2025 - present

Self Project (in progress) | **HTML, CSS, JavaScript** | [GitHub link](#)

- Building a **multi-stage web game** where players interact with customers, select cup sizes, and make drinks
- Developed **randomized order generation** with recipe structures for beverages and customization options
- Implemented **dialogue system** with alternating speech bubbles, timed interactions, and conditional UI
- Designed and created all **game assets from scratch**, including characters, backgrounds, icons and buttons

Studying Colour Gradient in Galaxies

Aug 2025 - Nov 2025

Course Project | Astrophysics | [report](#)

- Processed multi-band (g, r, i) optical data from the Growth India Telescope, performing **masking**, **sky background subtraction**, and **photometric zero-point calibration** using **Pan-STARRS** reference stars
- Conducted **isophotal fitting** and surface-brightness profile extraction using **Python (Astropy, Photutils, Autoprof, SciPy)** to derive radial colour profiles and measure **gradients in mag/dex**
- Quantified **inner and outer colour gradients** via **log-radius regression analysis**, interpreting **stellar population age**, **metallicity distribution**, and structural differences between **early- and late-type galaxies**

EDUCATION

B.Tech. in Engineering Physics | IIT Bombay

Oct 2022 - May 2026

Relevant Coursework

- **Computer Science** : Digital Image Processing of Remotely Sensed Data, Speech and Natural Language Processing and the Web, Computer Programming and Utilization, **AI and Data Science**
- **Mathematics and Physics** : Numerical Analysis, Calculus, Linear Algebra, Differential Equations, Non-linear Dynamics, Astrophysics, **Computational Many-Body Physics**
- **Design** : Design Thinking and Innovation, Pottery for Beginners